

Again, terms in each section need to be compared only if they have dashes in the same position. Note that the term $(000-)$ does not match with any other term. Therefore, it has no check mark at its right. The decimal equivalents are written on the left-hand side of each entry for identification purposes. The comparing process should be carried out again in column (c) and in subsequent columns as long as proper matching is encountered. In the present example, the operation stops at the third column.

[Step 4:] The unchecked terms in the table form the prime implicants. In this example we have the term $w'x'y'(000-)$ in column (b), and the terms $x'z'(-0-0)$ and $wy(1-1-)$ in column (c).

The sum of the prime-implicants gives a simplified expression for the function. This is because each checked term in the table has been taken into account by an entry of a simpler term in a subsequent column.

For the present example, the sum of prime-implicants gives the minimized function in sum of products:

$$F = w'x'y' + x'z' + wy$$

Map for the function of Example 3.13

w \ xz	00	01	11	10
00	1	1		1
01				
11			1	1
10	1		1	1

$$F = wy + x'z' + w'x'y'$$

The tedious manipulation that one must undergo when using the tabulation method is reduced if the comparing is done with decimal numbers instead of binary. A method will now be shown that uses subtraction of decimal numbers instead of the comparing and matching of binary numbers.

We note that each 1 in a binary number represents the coefficient multiplied by a power of 2. When two minterms are the same in every position except one, the minterm with the extra 1 must be larger than the